

Assignment Cover Sheet

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Assignment Title: System Modelling

Student Name(s) and Number(s)

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Date 28 August 2026

Statement of Collaboration:

This assignment was completed collaboratively by Jaylen Avtarovski, Elina Kim, and Rhyse Williams, with all three members contributing equally to the derivations, analysis, and written responses across Q1, Q2, and Q3. All group discussions, calculations, and write-ups were conducted jointly, and any external tools or resources used have been acknowledged in accordance with UTS academic integrity guidelines.

Contribution Weighting

Student Number	Student Name	Q1	Q2	Q3
24767939	Jaylen Avtarovski	100	100	100
24817532	Rhyse Williams	100	100	100
24949885	Elina Kim	100	100	100

Fill in each team member's percentage contribution to each question before submission.

Answers to the Assignment Questions:

Q1: Cart Pendulum Modelling

(20 marks)

Using the required generalised coordinates,

$$q_1 = x_c \quad q_2 = \phi$$

where x_c is the horizontal displacement of the cart and ϕ is measured anticlockwise from the negative x -axis.

From the geometry of the system, the pendulum mass position is

$$\begin{aligned} x_p &= x_c - l \cos \phi \\ y_p &= -l \sin \phi \end{aligned}$$

where l is the distance from the pivot to the centre of oscillation.

Differentiating with respect to time,

$$\begin{aligned} \dot{x}_p &= \dot{x}_c + l\dot{\phi} \sin \phi \\ \dot{y}_p &= -l\dot{\phi} \cos \phi \end{aligned}$$

The total co-kinetic energy of the system is the sum of the cart and pendulum kinetic energies:

$$T^* = T_c + T_p$$

with

$$T_c = \frac{1}{2} M \dot{x}_c^2$$

and

$$T_p = \frac{1}{2} m (\dot{x}_p^2 + \dot{y}_p^2)$$

Therefore,

$$T^* = \frac{1}{2} M \dot{x}_c^2 + \frac{1}{2} m \left[(\dot{x}_c + l\dot{\phi} \sin \phi)^2 + (-l\dot{\phi} \cos \phi)^2 \right]$$

Simplifying,

$$\begin{aligned} T^* &= \frac{1}{2} M \dot{x}_c^2 + \frac{1}{2} m (\dot{x}_c^2 + 2l\dot{x}_c\dot{\phi} \sin \phi + l^2\dot{\phi}^2) \\ T^* &= \frac{1}{2} (M + m) \dot{x}_c^2 + ml\dot{x}_c\dot{\phi} \sin \phi + \frac{1}{2} ml^2\dot{\phi}^2 \end{aligned}$$

Taking the pivot height as the gravitational reference,

$$\begin{aligned} V &= mg y_p \\ V &= -mgl \sin \phi \end{aligned}$$

The viscous friction of the cart and pendulum is represented using the Rayleigh dissipation function,

$$D = \frac{1}{2} b_c \dot{x}_c^2 + \frac{1}{2} b_p \dot{\phi}^2$$

The Lagrangian of the system is defined as

$$L = T^* - V$$

Substituting the co-kinetic and potential energy expressions gives

$$L = \frac{1}{2} (M + m) \dot{x}_c^2 + m \dot{x}_c \dot{\phi} \sin \phi + \frac{1}{2} m l^2 \dot{\phi}^2 + mgl \sin \phi$$

The Euler–Lagrange equation including viscous dissipation is

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} + \frac{\partial D}{\partial \dot{q}_i} = Q_i$$

For $q_1 = x_c$,

$$\frac{\partial L}{\partial \dot{x}_c} = (M + m) \dot{x}_c + m l \dot{\phi} \sin \phi$$

Taking the time derivative,

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_c} \right) = (M + m) \ddot{x}_c + m l \ddot{\phi} \sin \phi + m l \dot{\phi}^2 \cos \phi$$

Also,

$$\frac{\partial L}{\partial x_c} = 0$$

and

$$\frac{\partial D}{\partial \dot{x}_c} = b_c \dot{x}_c$$

Since the generalised force associated with x_c is

$$Q_{x_c} = F$$

substitution into the Euler–Lagrange equation gives the cart equation

$$\boxed{F = (M + m) \ddot{x}_c + b_c \dot{x}_c + m l \ddot{\phi} \sin \phi + m l \dot{\phi}^2 \cos \phi} \quad (1a)$$

Using the same Euler–Lagrange equation:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) - \frac{\partial L}{\partial \phi} + \frac{\partial D}{\partial \dot{\phi}} = Q_\phi$$

For this system,

$$Q_\phi = 0$$

Considering the pendulum coordinate $q_2 = \phi$,

$$\frac{\partial L}{\partial \dot{\phi}} = m\dot{x}_c \sin \phi + ml^2 \dot{\phi}$$

Taking the time derivative,

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = m\ddot{x}_c \sin \phi + m\dot{x}_c \dot{\phi} \cos \phi + ml^2 \ddot{\phi}$$

Now differentiate L with respect to ϕ :

$$\frac{\partial L}{\partial \phi} = m\dot{x}_c \dot{\phi} \cos \phi + mgl \cos \phi$$

For the dissipation term,

$$\frac{\partial D}{\partial \dot{\phi}} = b_p \dot{\phi}$$

Substituting into Euler–Lagrange:

$$m\ddot{x}_c \sin \phi + m\dot{x}_c \dot{\phi} \cos \phi + ml^2 \ddot{\phi} - \left(m\dot{x}_c \dot{\phi} \cos \phi + mgl \cos \phi \right) + b_p \dot{\phi} = 0$$

The matching cross terms cancel, leaving

$$\boxed{ml^2 \ddot{\phi} + b_p \dot{\phi} + m\ddot{x}_c \sin \phi - mgl \cos \phi = 0} \quad (1b)$$

Q2: Analytical Linear Model of a Cart Pendulum System

(40 marks)

The nonlinear model provided:

$$K_v \cdot u = (M + m) \ddot{x}_c + b_c \dot{x}_c + m\ell \ddot{\alpha} \cos \alpha - m\ell \dot{\alpha}^2 \sin \alpha \quad (1a)$$

$$0 = m\ell^2 \ddot{\alpha} + b_p \dot{\alpha} + m\ell \ddot{x}_c \cos \alpha + gml \sin \alpha \quad (1b)$$

Q2.1

The system is linearised around $\alpha = 0$ corresponding to the pendulum vertically facing downwards.

For small angular displacements around $\alpha = 0$,

$$\sin \alpha \approx \alpha$$

$$\cos \alpha \approx 1$$

The nonlinear higher-order term is neglected:

$$\dot{\alpha}^2 \sin \alpha \approx 0$$

Applying approximations to (1a):

$$K_v u = (M + m) \ddot{x}_c + b_c \dot{x}_c + m\ell \ddot{\alpha}(1) - m\ell \dot{\alpha}^2(0)$$

Applying approximations to (1b):

$$0 = m\ell^2 \ddot{\alpha} + b_p \dot{\alpha} + m\ell \ddot{x}_c(1) + gml(\alpha)$$

Therefore,

$$K_v u = (M + m) \ddot{x}_c + b_c \dot{x}_c + m\ell \ddot{\alpha} \quad (2a)$$

$$0 = m\ell^2 \ddot{\alpha} + b_p \dot{\alpha} + m\ell \ddot{x}_c + gml\alpha \quad (2b)$$

Q2.2

Assuming zero initial conditions, applying Laplace transform to (2a):

$$K_v U(s) = (M + m) s^2 X_c(s) + b_c s X_c(s) + m \ell s^2 A(s)$$

Taking Laplace transform of (2b):

$$0 = m \ell^2 s^2 A(s) + b_p s A(s) + m \ell s^2 X_c(s) + g m \ell A(s)$$

Therefore,

$$K_v U(s) = [(M + m) s^2 + b_c s] X_c(s) + m \ell s^2 A(s) \quad (3a)$$

$$0 = m \ell s^2 X(s) + [m \ell^2 s^2 + b_p s + g m \ell] A(s) \quad (3b)$$

$G_{x\alpha}(s)$:

Rearranging (3b),

$$[m \ell^2 s^2 + b_p s + g m \ell] A(s) = -m \ell s^2 X_c(s)$$

$$\frac{A(s)}{X_c(s)} = \frac{-m \ell s^2}{m \ell^2 s^2 + b_p s + g m \ell}$$

Therefore,

$$G_{x\alpha}(s) = \frac{A(s)}{X_c(s)} = -\frac{m \ell s^2}{m \ell^2 s^2 + b_p s + g m \ell} \quad (4a)$$

$$G_{x\alpha}(s) = -\frac{s^2}{\ell s^2 + \frac{b_p}{m \ell} s + g} \quad (4b)$$

$G_{ux}(s)$:

From (4a),

$$A(s) = -\frac{m \ell s^2}{m \ell^2 s^2 + b_p s + g m \ell} X_c(s)$$

Substituting into (3a),

$$K_v U(s) = [(M + m) s^2 + b_c s] X_c(s) + m \ell s^2 \left[-\frac{m \ell s^2}{m \ell^2 s^2 + b_p s + g m \ell} X_c(s) \right]$$

$$K_v U(s) = \left[(M + m) s^2 + b_c s - \frac{m^2 \ell^2 s^4}{m \ell^2 s^2 + b_p s + g m \ell} \right] X_c(s)$$

Therefore,

$$\frac{X_c(s)}{U(s)} = \frac{K_v}{(M + m) s^2 + b_c s - \frac{m^2 \ell^2 s^4}{m \ell^2 s^2 + b_p s + g m \ell}}$$

Multiplying the numerator and denominator by $m \ell^2 s^2 + b_p s + g m \ell$,

$$G_{ux}(s) = \frac{K_v (m \ell^2 s^2 + b_p s + g m \ell)}{[(M + m) s^2 + b_c s] [m \ell^2 s^2 + b_p s + g m \ell] - m^2 \ell^2 s^4}$$

Expanding the denominator,

$$\begin{aligned} & [(M + m) s^2 + b_c s] [m \ell^2 s^2 + b_p s + g m \ell] - m^2 \ell^2 s^4 \\ &= (M + m) m \ell^2 s^4 + (M + m) b_p s^3 + (M + m) g m \ell s^2 + b_c m \ell^2 s^3 + b_c b_p s^2 + b_c g m \ell s - m^2 \ell^2 s^4 \\ &= M m \ell^2 s^4 + [(M + m) b_p + b_c m \ell^2] s^3 + [(M + m) g m \ell + b_c b_p] s^2 + b_c g m \ell s \end{aligned}$$

Therefore,

$$G_{ux}(s) = \frac{K_v (m \ell^2 s^2 + b_p s + g m \ell)}{M m \ell^2 s^4 + [(M + m) b_p + b_c m \ell^2] s^3 + [(M + m) g m \ell + b_c b_p] s^2 + b_c g m \ell s} \quad (5)$$

$G_{u\alpha}(s)$:

The overall cascaded transfer function is,

$$G_{u\alpha}(s) = G_{ux}(s) G_{x\alpha}(s)$$

Substituting (4a) and (5),

$$G_{u\alpha}(s) = \frac{K_v (m \ell^2 s^2 + b_p s + g m \ell)}{M m \ell^2 s^4 + [(M + m) b_p + b_c m \ell^2] s^3 + [(M + m) g m \ell + b_c b_p] s^2 + b_c g m \ell s} \left[\frac{-m \ell s^2}{m \ell^2 s^2 + b_p s + g m \ell} \right]$$

Therefore,

$$G_{u\alpha}(s) = \frac{-K_v m \ell s^2}{M m \ell^2 s^4 + [(M + m) b_p + b_c m \ell^2] s^3 + [(M + m) g m \ell + b_c b_p] s^2 + b_c g m \ell s} \quad (6)$$

Q2.3

From (1a), the effect of the pendulum angle over the cart is defined as,

$$d = \frac{m \ell}{K_v} (\ddot{\alpha} \cos \alpha - \dot{\alpha}^2 \sin \alpha)$$

Thus,

$$u = \frac{1}{K_v} [(M + m) \ddot{x}_c + b_c \dot{x}_c] + d$$

For small values of m , ℓ and small angular oscillations,

$$d \approx 0$$

Therefore,

$$\begin{aligned} u &= \frac{1}{K_v} [(M + m) \ddot{x}_c + b_c \dot{x}_c] \\ K_v u &= (M + m) \ddot{x}_c + b_c \dot{x}_c \end{aligned}$$

Assuming zero initial conditions and applying the Laplace transform,

$$\begin{aligned} K_v U(s) &= (M + m) s^2 X_c(s) + b_c s X_c(s) \\ &= [(M + m) s^2 + b_c s] X_c(s) \end{aligned}$$

Therefore,

$$\frac{X_c(s)}{U(s)} = \frac{K_v}{(M + m)s^2 + b_c s}$$

Hence,

$$\tilde{G}_{ux}(s) = \frac{K_v}{(M + m)s^2 + b_c s} \quad (7a)$$

$$\tilde{G}_{ux}(s) = \frac{K_v}{s[(M + m)s + b_c]} \quad (7b)$$

Q3: Data-Based System Model Identification

(40 marks)

Known parameters: $M = 0.22$ kg, $m = 0.1$ kg, $g = 9.81$ m/s². Unknown: b_c , b_p , ℓ and K_v .

Note: the supplied `main_Q33_Q34.m` initialises $g = 8.1$; this has been corrected to $g = 9.81$ as specified in the assignment brief, since ℓ scales linearly with g through (4).

Q3.1) Identification of $\tilde{G}_{ux}(s)$

Following Remark 2, the identification is carried out on the *voltage-to-speed* channel and the integrator is appended afterwards, which avoids fitting the marginally stable pole at $s = 0$.

From (??) the model structure to be fitted is first order with no zeros, so in `main_Q31_Q32.m` the input/output signals and model orders must be set to

<code>y = vc;</code>	TF output – filtered derivative of <code>xc_m</code>
<code>u = Vm;</code>	TF input – measured motor voltage
<code>np = 1;</code>	one pole (from $s + b$)
<code>nz = 0;</code>	no zeros

so that $G_{uvc_id} = \hat{a}/(s + \hat{b})$ and $G_{ux_id} = \frac{1}{s}G_{uvc_id}$. Record the fitted coefficients as

$$\hat{G}_{uvc}(s) = \frac{\hat{a}}{s + \hat{b}} \implies \tilde{G}_{ux}(s) = \frac{\hat{a}}{s(s + \hat{b})}. \quad (1)$$

Numerical result (from `main_Q31_Q32.m`): $\hat{a} = 0.887$, $\hat{b} = 16.25$, fit to estimation data = 93.97%.

Q3.2) Identification of $G_{x\alpha}(s)$

Following Remark 3, the fit is performed between the cart acceleration $a_c(t) = \ddot{x}_c(t)$ and the angle $\alpha(t)$. From (??) the correct structure is second order with no zeros:

<code>y = alpha_rad;</code>	TF output – measured angle in radians
<code>u = ac;</code>	TF input – filtered cart acceleration
<code>np = 2;</code>	two poles (pendulum resonance)
<code>nz = 0;</code>	no zeros

Normalising the estimate returned by `tfest` so that the leading denominator coefficient is unity,

$$\hat{G}_{ac\alpha}(s) = \frac{\hat{k}}{s^2 + \hat{a}_1 s + \hat{a}_0}, \implies \hat{G}_{x\alpha}(s) = s^2 \hat{G}_{ac\alpha}(s) = \frac{\hat{k} s^2}{s^2 + \hat{a}_1 s + \hat{a}_0}. \quad (2)$$

Numerical result (from `main_Q31_Q32.m`): $\hat{k} = -3.902$, $\hat{a}_1 = 0.5832$, $\hat{a}_0 = 38.58$, fit to estimation data = 78.4%.

Q3.3) Estimation of b_c , b_p , ℓ and K_v

The unknown parameters follow from a coefficient-by-coefficient comparison of the identified transfer functions with the analytical ones derived in Q2.

Step 1 – from $G_{x\alpha}(s)$: obtain ℓ and b_p (Remark 4). Comparing (2) with the analytical expression (??),

$$\frac{\hat{k}}{s^2 + \hat{a}_1 s + \hat{a}_0} \equiv \frac{-1/\ell}{s^2 + \frac{b_p}{m\ell^2}s + \frac{g}{\ell}} \Rightarrow \begin{cases} \hat{a}_0 = \frac{g}{\ell}, \\ \hat{a}_1 = \frac{b_p}{m\ell^2}, \\ \hat{k} = -\frac{1}{\ell}. \end{cases} \quad (3)$$

The first relation gives the length directly, the second the pivot friction:

$$\ell = \frac{g}{\hat{a}_0} = \frac{g}{\hat{\omega}_n^2}, \quad \hat{\omega}_n = \sqrt{\hat{a}_0} \quad (4)$$

$$b_p = m\ell^2 \hat{a}_1 = \frac{m g^2 \hat{a}_1}{\hat{a}_0^2} = 2\zeta \hat{\omega}_n m \ell^2, \quad \zeta = \frac{\hat{a}_1}{2\sqrt{\hat{a}_0}} \quad (5)$$

$$L = \frac{3}{2} \ell \quad (\text{since } \ell = 2L/3) \quad (6)$$

The third relation, $\hat{k} = -1/\ell$, is redundant and provides a useful **consistency check**: the value $-1/\hat{k}$ should agree with $g/\hat{a}_0$ to within the identification error. (If `tfest` returns $\hat{k} > 0$, the encoder sign convention for α is inverted with respect to Fig. 2; negate `alpha_rad` and re-identify.)

Step 2 – from $\tilde{G}_{ux}(s)$: obtain K_v and b_c (Remark 4). Comparing (1) with the analytical form (??), $a = K_v/(M + m)$ and $b = b_c/(M + m)$, hence

$$K_v = (M + m) \hat{a} = 0.32 \hat{a} \quad [\text{N/V}] \quad (7)$$

$$b_c = (M + m) \hat{b} = 0.32 \hat{b} \quad [\text{N s/m}] \quad (8)$$

Equivalently, in terms of the steady-state gain and time constant of the first-order speed response, $K_v = b_c \hat{a}/\hat{b}$ and $b_c = (M + m)/\hat{\tau}$.

Summary of identified parameters.

Parameter	Expression	Value	Units
ℓ	$g/\hat{a}_0$	0.2542768274	m
L	$3\ell/2$	0.3814152411	m
b_p	$m\ell^2 \hat{a}_1$	0.003770779033	N m s/rad
K_v	$(M + m) \hat{a}$	0.28384	N/V
b_c	$(M + m) \hat{b}$	5.2	N s/m

Sanity checks. Before accepting the numbers, verify that (i) ℓ is of the order of two thirds of the physical rod length measured in the lab; (ii) $\zeta \ll 1$, consistent with the lightly damped oscillation visible in Fig. 5; and (iii) the observed oscillation period in Fig. 5 satisfies $T \approx 2\pi\sqrt{\ell/g}$.

Q3.4) Nonlinear and linear simulation models

The Simulink file `sim_2026_cart_pend.slx` implements the two models below with the parameters identified in Q3.3.

Nonlinear model. Writing (1a)–(1b) in matrix form,

$$\begin{bmatrix} M + m & m\ell \cos \alpha \\ m\ell \cos \alpha & m\ell^2 \end{bmatrix} \begin{bmatrix} \ddot{x}_c \\ \ddot{\alpha} \end{bmatrix} = \begin{bmatrix} K_v u - b_c \dot{x}_c + m\ell \dot{\alpha}^2 \sin \alpha \\ -b_p \dot{\alpha} - mg\ell \sin \alpha \end{bmatrix}, \quad (9)$$

whose mass-matrix determinant is $\det = m\ell^2(M + m \sin^2 \alpha) > 0$ for all α . Inverting,

$$\ddot{x}_c = \frac{\ell(K_v u - b_c \dot{x}_c + m\ell \dot{\alpha}^2 \sin \alpha) + \cos \alpha(b_p \dot{\alpha} + mg\ell \sin \alpha)}{\ell(M + m \sin^2 \alpha)}, \quad (10)$$

$$\ddot{\alpha} = \frac{-(M + m)(b_p \dot{\alpha} + mg\ell \sin \alpha) - m\ell \cos \alpha(K_v u - b_c \dot{x}_c + m\ell \dot{\alpha}^2 \sin \alpha)}{m\ell^2(M + m \sin^2 \alpha)}. \quad (11)$$

Equations (10)–(11) are implemented with integrator blocks initialised at $x_c(0) = v_c(0) = \alpha(0) = \omega(0) = 0$.

Linear model. As required by Remark 6, the linear branch uses the **full** $G_{ux}(s)$ of (??)–(??) — not the simplified $\tilde{G}_{ux}(s)$ — cascaded with $G_{x\alpha}(s)$ of (??), exactly as in Fig. 3.

Comparison. *[Insert the plot generated at the end of main_Q33_Q34.m comparing $x_c(t)$ and $\alpha(t)$ from the non-linear and linear models for the same input $u(t)$.]*

The two responses are expected to be indistinguishable while $|\alpha| \lesssim 10^\circ$, and to diverge for larger excitations, where the neglected terms $m\ell \dot{\alpha}^2 \sin \alpha$ and the curvature of $\sin \alpha$ become significant — which quantifies the validity range of the linearisation assumptions A1)–A2) of Q2.1.

References

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